

Curated Results, Tables, and Figure Compendium

Companion to Manuscript v11 – matched multi-omic constraints on the spaceflight kidney
NCC/SPAK/WNK phospho-phenotype

Ibrahim Shahid

Draft v11 results compendium, May 2026

This document collects the curated results, manuscript tables, figure gallery, and artifact index supporting Manuscript v11. It is the technical supplement for the streamlined manuscript: cross-cohort RNA recurrence, matched OSD-462 RNA/protein/phosphoprotein mismatch, distal-nephron DCT-subtype-prior phosphosite enrichment, parent-protein and compartment-score sensitivity models, exploratory covariance decomposition, external spatial-reference context, and the public perturbation-reference pass (low-K DCT, parent-protein-normalized phosphosite effects, IRI DCT-transport spatial context, dDAVP/mpkDCT, KLHL3/CUL3 turnover). The main manuscript frames these analyses as a constraint map around the established spaceflight kidney NCC/SPAK/WNK phospho-phenotype; this compendium carries the extra tables, secondary analyses, and run artifacts.

Contents

S1	Run provenance and executable entrypoint	S2
S2	Executive verdict and claim labels	S3
S3	RNA recurrence and RNA-protein mismatch	S4
S4	GSE228367 DCT subtype prior	S4
S5	Distal-nephron subtype-prior phosphosite enrichment	S5
S6	Parent-protein and composition-score sensitivity	S7
S7	PXD001729 and KLHL3/CUL3 checks	S8
S8	Exploratory covariance decomposition	S8
S9	External spatial reference contextualization	S9
S10	Public perturbation-reference analyses	S11
	S10.1 Branch 1: GSE228367 low-K DCT anti-alignment	S12
	S10.2 Branch 2: parent-protein-normalized DCT-prior enrichment	S13
	S10.3 Branch 3: IRI Visium DCT-transport spatial context	S13
	S10.4 Branch 4: PXD001729 dDAVP/mpkDCT	S14
	S10.5 Branch 5: KLHL3/CUL3/WNK turnover	S14
S11	Human, aldosterone, and CMap appendix context	S14
S12	Complete artifact index	S15
S13	Proteome observability-bias audit	S17
S14	Verification notes	S18

S1 Run provenance and executable entriypoint

Table S1: V11 run provenance.

Run root	data/results/run_20260526_v11_dct1_phospho_mediation/
Plan	docs/v11_execution_research_plan.md
Execution report	docs/v11_execution_results.md
Plan checksum	d8b7465e09257dd4444fc2abb969a02066f2d360e4fbc04056791a60c9755e3f
Pipeline entriypoint	python -m src.run_all_phases --v11-only --run-id <run_id>
Canonical v11 modules	src/v11/build_gse228367_dct_prior.R; src/v11/core_analysis.py; src/v11/h2_composition_aware_phospho.py; src/v11/rna_protein_propagation.py; src/v11/observability_audit.py; src/v11/spatial_reference_projection.py; src/v11/publication_figures.py
Public perturbation-reference modules	src/v11/perturbation_gse228367_lowk.py; src/v11/h2_occupancy_normalized_phospho.py; src/v11/spatial_dct_transport_check.py; src/v11/perturbation_triage_summary.py

Table S2: External datasets added or reused for v11.

Layer	Dataset	V11 role
Spaceflight RNA recurrence	RRRM-2 / OSD-771, OSD-513, OSD-462	Defines and tests the recurrent matrix/endothelial-high and DCT-low RNA state.
Spaceflight multi-omics	OSD-462 / RR-10	Whole-kidney matched RNA, protein, and phosphoprotein anchor.
DCT subtype prior	GSE228367	Native kidney DCT1/DCT2 snRNA-seq reference used to score parent-gene DCT1-high and DCT2-leaning priors.
DCT-lineage phospho-proteome	PXD001729	mpkDCT dDAVP phosphoproteomic reference; used for DCT-lineage plausibility and negative anti-alignment boundary.
Spatial reference	GSE269622	Whole-transcriptome Visium IRI timecourse used for pathway-vector projection.
Spatial annotation context	GSE269719	Targeted 299-gene Xenium object used for cell-type and cellular-neighborhood inventory.

S2 Executive verdict and claim labels

Table S3: Main result verdicts.

Hypothesis	Question	Verdict
RNA recurrence	Does a recurrent matrix/endothelial-high and DCT-low RNA remodeling state recur across cohorts and in OSD-462 RNA?	Supported; remains the backbone.
DCT-subtype-prior enrichment	Are OSD-462 flight-suppressed phosphosites enriched among distal-nephron subtype-prior parent-gene extremes?	Supported for DCT1-high and DCT2-leaning decile bins; DCT1 is not exclusive.
Sensitivity models	Does the enrichment remain after parent-protein, parent-gene, observability, and compartment-score sensitivity checks?	Top-decile DCT1 subset survives many checks but attenuates in matched parent-gene permutation; DCT2-leaning is stronger in several parent-gene-aware summaries.
Exploratory covariance decomposition	Are remodeling scores statistically related to NCC/SPAK phospho suppression?	Endothelial and matrix/endothelial covariance summaries are directionally consistent; temporal order remains experimental.
Spatial supplement	Which injury/repair spatial niches resemble the bulk space-flight RNA vector?	Contextualized to repair-stage, DCT-adjacent/injured-tubule and fibro-inflammatory/endothelial niches.
Public perturbation references	Do public perturbation references (low-K DCT, parent-protein normalization, IRI DCT-transport, dDAVP, KLHL3/CUL3) identify the upstream WNK-suppressing mechanism?	No single upstream signal identified; parent-protein-normalized DCT-prior enrichment is a robustness upgrade; IRI Visium DCT-adjacent transport drop is a testable spatial prediction.

Table S4: Preferred and avoided manuscript language.

Topic	Preferred wording	Avoid
Overall frame	Distal-nephron subtype-prior phosphosite parent-gene enrichment within recurrent kidney remodeling.	DCT1-anchored mechanism without qualification.
DCT-prior enrichment	DCT1-high and DCT2-leaning decile enrichment of flight-suppressed whole-kidney phosphosites.	DCT1-specific phosphoproteomics or DCT1 exclusivity.
Composition models	Bulk sensitivity adjustment for parent protein abundance and estimated compartments.	Fully deconfounded or deconvolved cell-of-origin analysis.
Covariance decomposition	Remodeling scores covary with NCC regulatory phosphorylation after conditioning choices.	Endothelial remodeling causes NCC dephosphorylation.
Spatial	External IRI spatial reference contextualization.	Spatial validation of the spaceflight lesion.

S3 RNA recurrence and RNA-protein mismatch

Table S5: Core RNA recurrence and multi-omic quantities from the v11 run.

Component	Value	Interpretation
OSD-462 RNA recurrence	0.869	OSD-462 recurs the RRRM-2 matrix-high / DCT-low direction.
OSD-462 protein abundance	null	RNA remodeling does not propagate cleanly as protein abundance.
NCC regulatory phospho score	-0.832	NCC regulatory phosphosites are suppressed while total NCC protein is flat.
Targeted KSEA SPAK/OSR1	-6.31	Inferred kinase-output down; three quantified curated substrates.
Targeted KSEA WNK	-4.12	Inferred WNK-output down; three quantified curated substrates, interpreted as pathway coherence.
Kinome-wide atlas SPAK/OSR1	+1.44/ +1.57	Slightly positive and non-significant in the unbiased substrate atlas; discordant with targeted SPAK/OSR1 KSEA.
Endothelial score vs NCC phospho	-0.762	Strongest compartment anti-correlation with NCC regulatory phosphorylation.

Interpretation: recurrent RNA remodeling is strong, targeted protein-abundance concordance is null, and the transporter lesion resolves at the phospho/activity layer.

Table S6: **OSD-462 cross-layer pathway matrix.** RNA effects are curated pathway-score flight effects. Protein effects are mean gene-level abundance effects over matched pathway genes. Phosphosite effects are parent-gene-level mean phosphosite effects, averaged over matched pathway genes so multi-site genes do not dominate the pathway summary. Direction calls are descriptive thresholds, not individual-feature significance tests. The final column gives the matched-null-calibrated layer assignment from `src/v11/rna_protein_propagation.py` (two pathways calibrated protein-inverted, three RNA→phospho carry candidates, six RNA-not-propagated within the abundance×peptide-count null); see Section S13 for the detectability robustness of these assignments.

Pathway	RNA	Protein	Phospho	Pattern	Layer assignment (calibrated)
ECM organization	+0.222	−0.107	−0.011	RNA–protein opposite	protein-inverted (calibrated)
MMP/ADAM proteolysis	+0.197	+0.012	−0.289	protein-flat; phosphosite down	RNA not propagated
Preservation/stress response	+0.157	−0.014	−0.024	protein-flat; phosphosite down	RNA not propagated
Fibrosis/TGFβ/EMT	+0.149	−0.060	−0.034	RNA–protein opposite	RNA not propagated
S1P/S1PR3	+0.123	−0.045	+0.079	RNA–protein opposite	RNA→phospho candidate
Integrin/cell adhesion	+0.119	+0.005	+0.022	protein-flat; phosphosite up	RNA not propagated
Macrophage inflammation	+0.109	−0.070	−0.154	RNA–protein opposite	RNA not propagated
TLR4/innate	+0.075	−0.034	+0.000	RNA–protein opposite	protein-inverted (calibrated)
Oxidative stress/NRF2	+0.021	−0.010	+0.021	RNA–protein concordant/near-flat	RNA not propagated
Broad tubular transport	−0.048	+0.044	−0.242	RNA–protein opposite	RNA→phospho candidate
DCT/NCC-WNK transport	−0.130	+0.057	−0.119	RNA–protein opposite; phosphosite down	RNA→phospho candidate

S4 GSE228367 DCT subtype prior

Table S7: GSE228367 DCT reference inventory.

DCT1 nuclei	18,881
DCT2 nuclei	6,274
Normal-potassium replicates per subtype	3
Strict FDR DCT1 core genes	2
Strict FDR DCT2 core genes	0

Table S8: **Selected marker sanity checks in the DCT1/DCT2 reference.**

Gene	DCT1 mean	DCT2 mean	Read
<i>Slc12a3</i>	5.14	4.40	present in both, higher in DCT1
<i>Pvalb</i>	0.279	0.024	DCT1-high
<i>Trpm6</i>	2.40	1.90	DCT1-high
<i>Khlh3</i>	2.84	2.65	mild DCT1-high
<i>Trpv5</i>	0.006	0.711	DCT2-high
<i>Calb1</i>	1.01	2.97	DCT2-high
<i>Stk39</i>	1.98	1.54	DCT1-leaning
<i>Wnk4</i>	1.72	1.40	DCT1-leaning

Interpretation: the strict FDR marker sets are too sparse for binary enrichment. V11 therefore uses continuous DCT1/DCT2 subtype-prior scores and percentile bins. Leave-one-replicate-out mean-difference priors were stable for the DCT1-high and DCT2-leaning decile bins (Jaccard approximately 0.85–0.86), while log-ratio and rank-average score definitions changed bin membership more strongly.

S5 Distal-nephron subtype-prior phosphosite enrichment

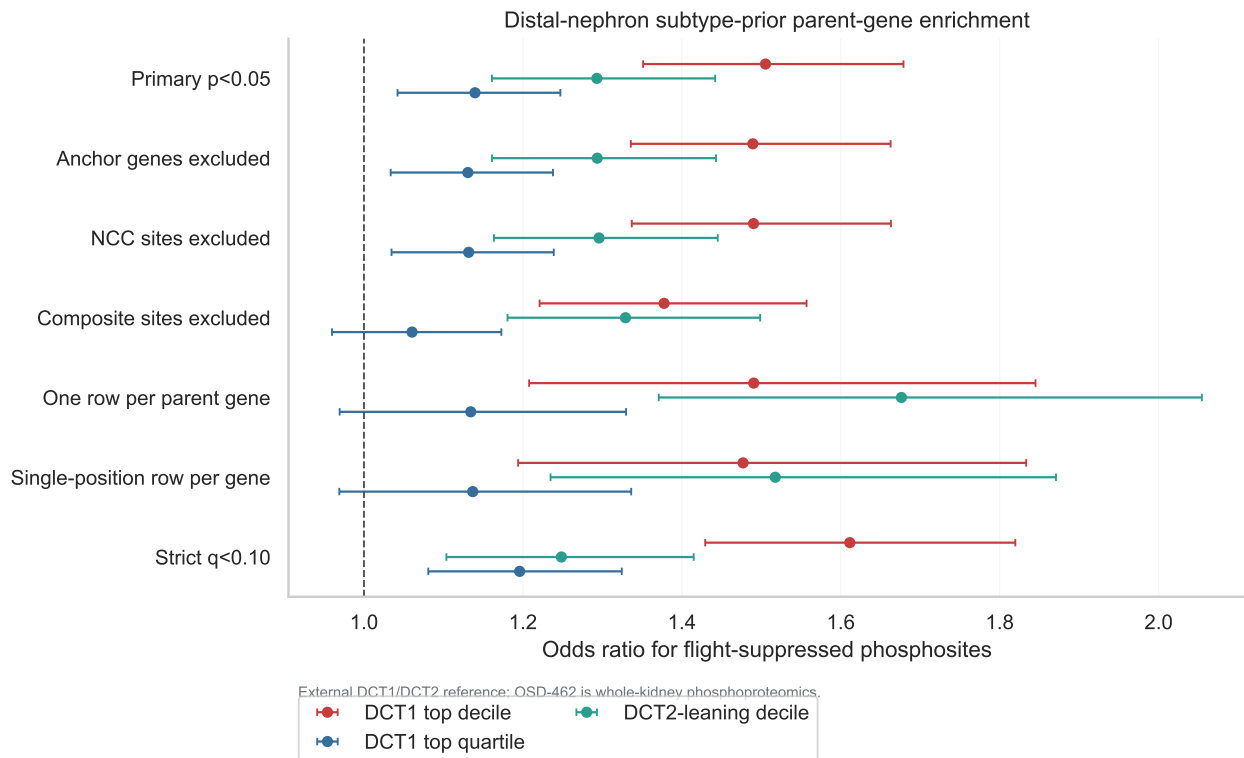


Figure S1: **Distal-nephron subtype-prior parent-gene enrichment.** DCT1-high, DCT2-leaning, and DCT1 top-quartile reference bins are tested across primary and sensitivity definitions. The safest claim is enrichment in distal-nephron subtype-prior extremes rather than DCT1 exclusivity.

Table S9: **Primary DCT1 top-decile enrichment tests within the DCT-subtype-prior analysis.**

Test	Suppressed set	Result	q	Read
Matched-null mean DCT1 score	pj0.05 suppressed sites	–	0.065	weak continuous support only
Fisher DCT1 top quartile	pj0.05 suppressed sites	OR 1.14	0.0061	supported but weaker than top decile
Fisher DCT1 top decile	pj0.05 suppressed sites	OR 1.51	1.33×10^{-11}	supported at row level
DCT1 top decile, anchor genes excluded	pj0.05 suppressed sites	OR 1.49	2.95×10^{-11}	survives anchor exclusion
DCT1 top decile, NCC sites excluded	pj0.05 suppressed sites	OR 1.49	2.95×10^{-11}	survives NCC-site exclusion
Composite/multi-position sites excluded	pj0.05 suppressed sites	OR 1.38	2.0×10^{-6}	removes composite rows; not a parent-gene dependence fix
One row per parent gene	pj0.05 suppressed sites	OR 1.49	6.59×10^{-4}	survives parent-gene representative-row sensitivity
Single-position row per parent gene	pj0.05 suppressed sites	OR 1.48	9.05×10^{-4}	stricter representative-row sensitivity
Parent-gene Fisher	parent has at least one pj0.05 suppressed site	OR 1.52	1.90×10^{-4}	passes parent-gene-level confirmation
Parent-gene logistic	parent has at least one pj0.05 suppressed site	OR 1.14	0.424	attenuated after site-count, peptide-count, abundance, and missingness
Matched parent-gene permutation	parent has at least one pj0.05 suppressed site	OR 1.51	$p_{\text{emp}} = 0.107$	covariates attenuated after observability matching
Cluster bootstrap	row-level pj0.05 suppressed sites	OR 1.51	1.22–1.81	parent-gene cluster-bootstrap interval excludes one

Table S10: **DCT2-leaning comparator across the same sensitivity ladder.** The DCT2-leaning bin contains the 10% of detected parent genes with the lowest DCT1 enrichment score. Both DCT1-high and DCT2-leaning deciles use the same one-sided Fisher exact alternative. In several parent-gene-aware summaries the DCT2-leaning odds ratio is larger than the DCT1-high odds ratio.

Analysis	DCT1 OR	DCT1 q	DCT2 OR	DCT2 q	Read
Primary $p < 0.05$	1.51	1.33×10^{-11}	1.29	1.30×10^{-5}	DCT1 stronger
Anchor genes excluded	1.49	2.95×10^{-11}	1.29	1.30×10^{-5}	DCT1 stronger
NCC sites excluded	1.49	2.95×10^{-11}	1.30	1.30×10^{-5}	DCT1 stronger
Composite rows excluded	1.38	2.0×10^{-6}	1.33	1.30×10^{-5}	similar
One row per parent gene	1.49	6.59×10^{-4}	1.68	4.62×10^{-6}	DCT2 larger
Single-position row per gene	1.48	9.05×10^{-4}	1.52	2.91×10^{-4}	both pass
Parent-gene Fisher	1.52	1.90×10^{-4}	1.82	3.29×10^{-8}	DCT2 larger
Matched parent-gene permutation	1.51	0.107	1.70	0.017	DCT1 attenuates

Interpretation: suppressed whole-kidney OSD-462 phosphosites are enriched among parent genes in distal-nephron subtype-prior extremes. The DCT1 top decile is biologically motivated by NCC/SPAK/WNK anchors, while the DCT2-leaning comparator is also enriched and is stronger in several parent-gene-aware summaries. The supported claim is therefore distal-nephron subtype-prior enrichment rather than DCT1 exclusivity. Direct cell-of-origin assignment would require DCT-enriched or spatially resolved spaceflight phosphoproteomics.

S6 Parent-protein and composition-score sensitivity

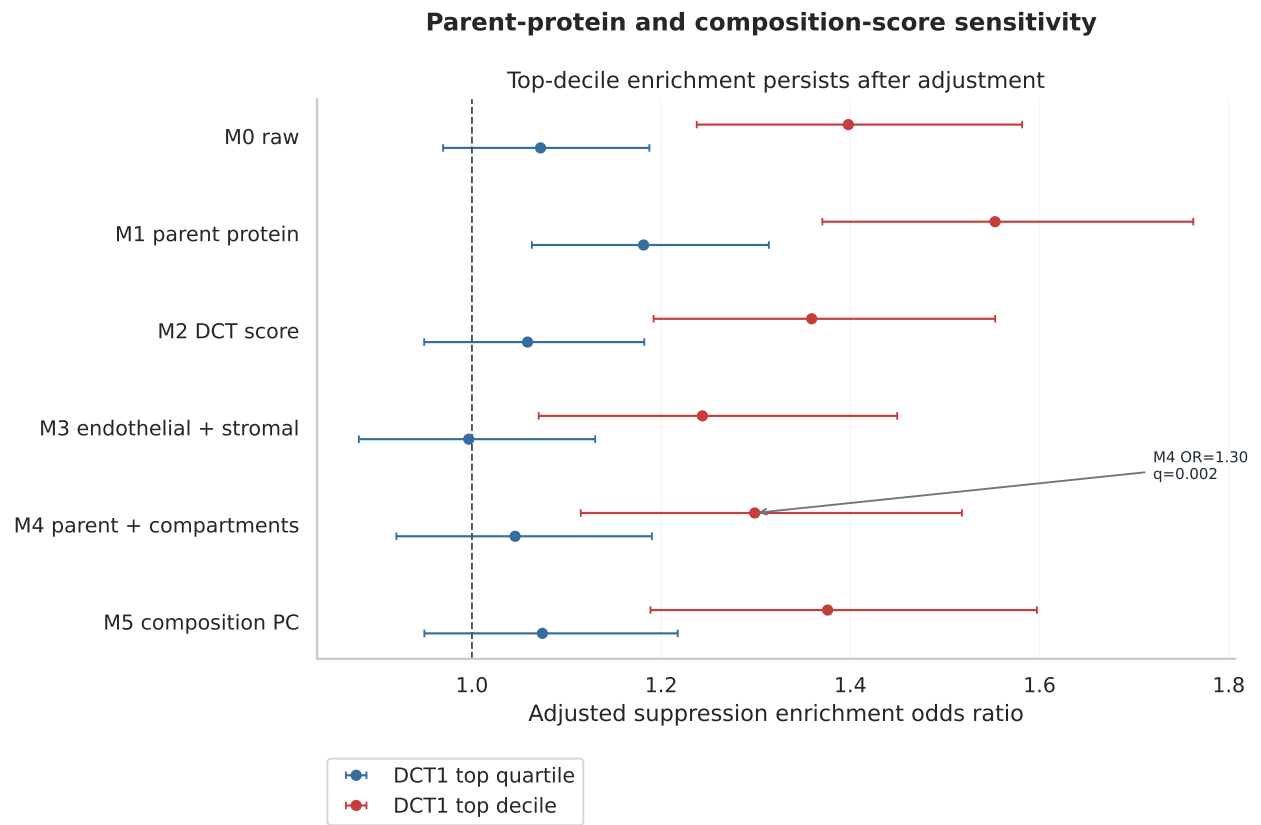


Figure S2: **Parent-protein and composition-score sensitivity.** Panel A shows top-decile and top-quartile enrichment across the adjustment ladder. Panel B shows that continuous DCT1-reference and flight-by-DCT1 interaction models remain weak.

Table S11: **DCT1 top-decile adjusted-suppression enrichment across the composition-aware model ladder.**

Model	Adjustment	OR	q	Read
M0	Raw flight effect	1.40	6.35e-7	raw DCT1 top-decile subset enrichment
M1	Parent protein abundance	1.55	2.14e-10	not eliminated by parent protein abundance adjustment
M2	DCT score	1.36	2.17e-5	survives estimated DCT abundance adjustment
M3	Endothelial + stromal scores	1.24	0.00545	attenuated, still positive
M4	Parent protein + DCT + endothelial + stromal	1.30	0.00158	full model remains positive
M5	Parent protein + composition PC1	1.38	6.53e-5	PC-based composition adjustment passes

Table S12: **Continuous DCT1-reference models do not pass.**

Test	Key term	Coefficient	q	Read
Two-stage M0 DCT1-only	DCT1 prior	-0.00110	0.877	weak negative, not significant
Two-stage M4 full	DCT1 prior	+0.00274	0.877	no continuous suppression gradient
One-shot LM0 site-fixed	flight x DCT1 prior	-0.00145	0.665	direction negative, weak
One-shot LM4 full site-fixed	flight x DCT1 prior	-0.00200	0.665	direction negative, weak

Interpretation: the robust result is top-decile subset enrichment after parent-protein and compartment-score adjustment, not a broad continuous DCT1-gradient effect. The adjusted models redefine the suppressed set on each residual scale (M0: 2,430 nominally suppressed single-position sites; M4: 1,390). A fixed-M0-set sensitivity showed that 99.3% of M0 suppressed sites with M4 covariates remained negative in the full model (mean adjusted effect -0.295), including 99.7% of DCT1 top-decile sites (mean -0.303). Sample-level covariate diagnostics showed endothelial–stromal correlation $r = 0.86$ and stromal-score VIF 6.59, so these are bulk sensitivity models rather than deconfounding.

S7 PXD001729 and KLHL3/CUL3 checks

The PXD001729 dDAVP/mpkDCT anti-alignment check and the KLHL3/CUL3/WNK turnover coverage check are presented under the public-perturbation-reference section (§S10 below), as Branches 4 and 5 of the triangulation pass. See Tables S23 and S24 for the shared-site cosine and the OSD-462/PXD coverage numbers, and the corresponding interpretation paragraphs.

S8 Exploratory covariance decomposition

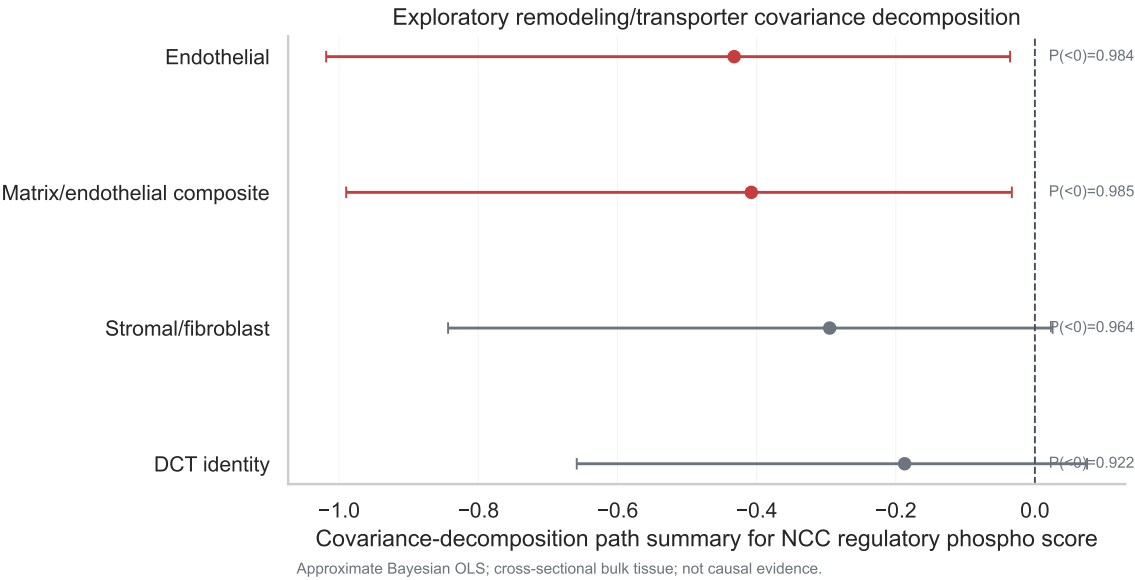


Figure S3: **Exploratory remodeling/transporter covariance decomposition.** Endothelial and matrix/endothelial composite summaries are negative with intervals excluding zero under the implemented conditioning choices. Cross-sectional bulk data prevent causal interpretation.

Table S13: **Covariance-decomposition summaries from approximate Bayesian OLS models.**

Intermediate score	Summary median	95% low	95% high	Read	
Endothelial	-0.432	-1.019	-0.036	negative phosphorylation summary	remodeling-covariance
Stromal/fibroblast	-0.295	-0.843	0.025	suggestive, interval crosses zero	
DCT identity	-0.187	-0.658	0.075	composition-linked, interval crosses zero	interval
Matrix/endothelial composite	-0.408	-0.990	-0.033	negative phosphorylation summary	remodeling-covariance

Table S14: **Approximate future-n planning from observed effect scale.**

Mediator	Approximate n for 80% interval-exclusion power
Endothelial	40 animals
Matrix/endothelial composite	40 animals
Stromal/fibroblast	60 animals
DCT identity	80 animals

Interpretation: the data are consistent with remodeling-linked covariance or shared remodeling, not causal evidence. The direct flight effect remains negative after intermediate-score adjustment.

S9 External spatial reference contextualization

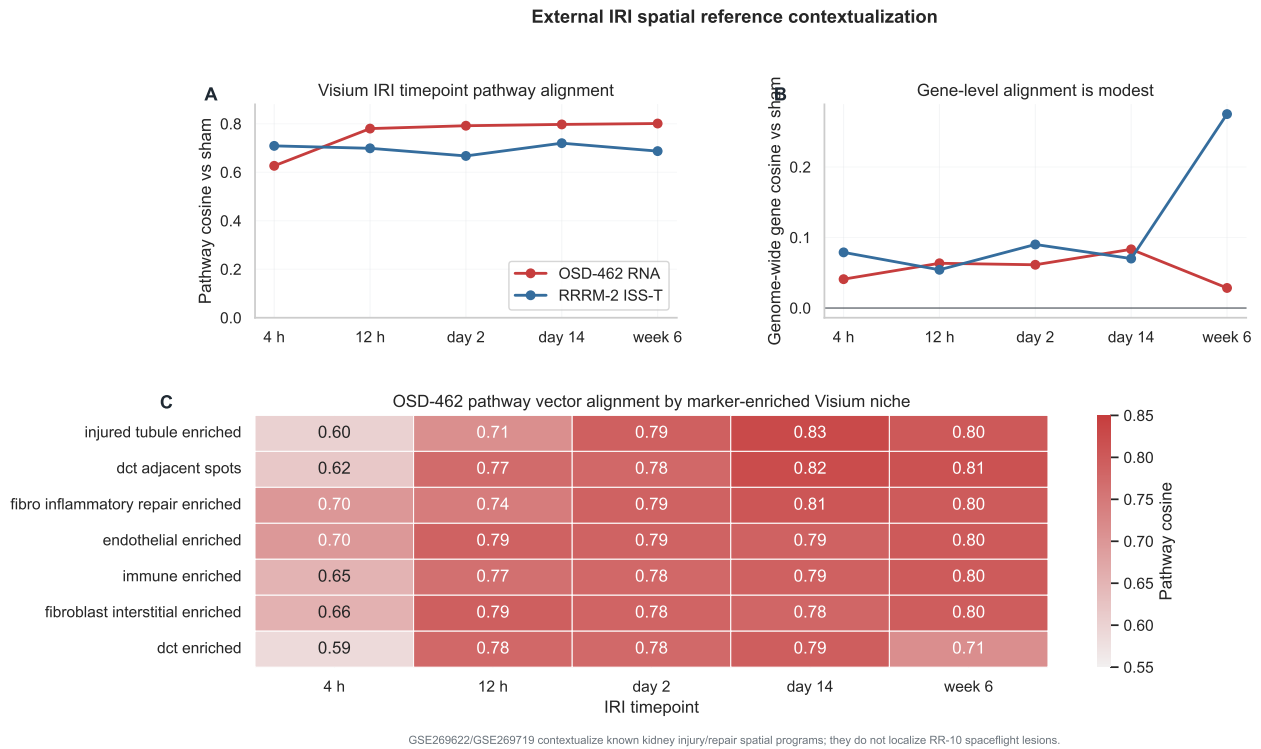


Figure S4: **External IRI spatial reference projection.** The OSD-462 RNA vector aligns strongly with Visium IRI repair-stage pathway programs. Spatial-niche matches are strongest in injured tubule, DCT-adjacent, fibro-inflammatory, and endothelial contexts.

Table S15: **Visium IRI timepoint cosines vs sham.**

IRI timepoint	OSD-462 pathway	RRRM-2 pathway	OSD-462 gene	RRRM-2 gene
4 h	0.627	0.709	0.041	0.079
12 h	0.780	0.699	0.063	0.054
day 2	0.792	0.667	0.061	0.090
day 14	0.797	0.720	0.083	0.070
week 6	0.801	0.687	0.028	0.275

Table S16: **Top OSD-462 Visium niche pathway matches.**

Timepoint	Niche	Cosine
day 14	injured tubule-enriched spots	0.827
day 14	DCT-adjacent spots	0.825
week 6	DCT-adjacent spots	0.814
day 14	fibro-inflammatory repair-enriched spots	0.807
week 6	injured tubule-enriched spots	0.802
week 6	endothelial-enriched spots	0.802

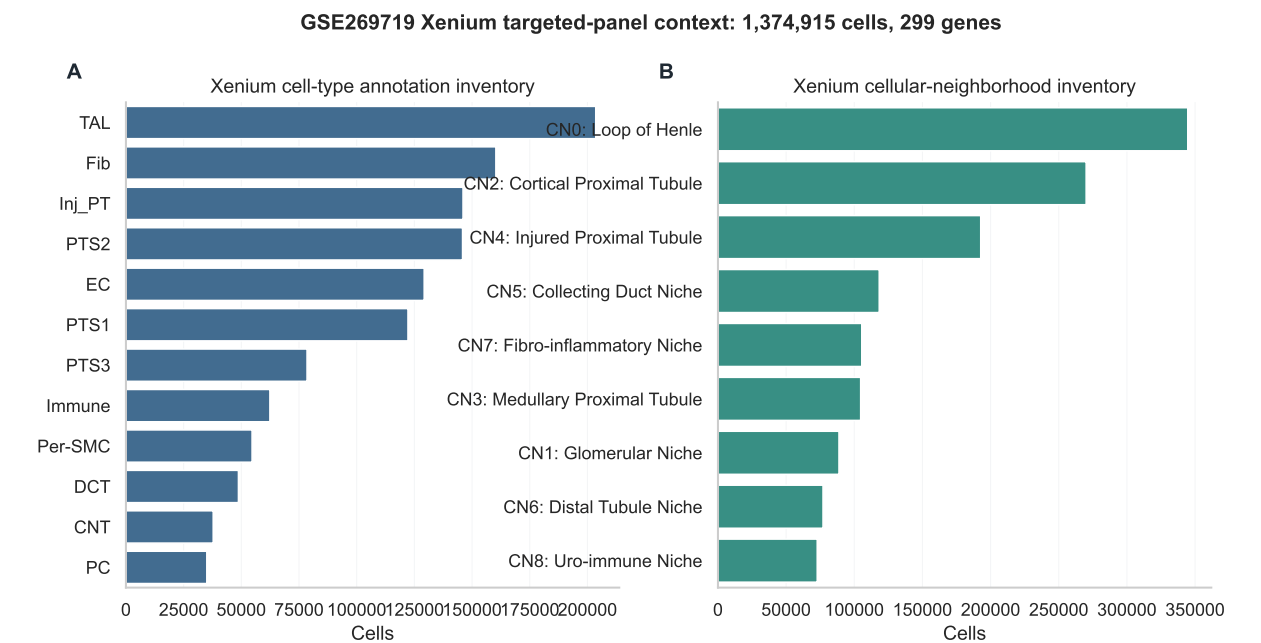


Figure S5: **GSE269719 Xenium annotation inventory.** The Xenium object provides targeted-panel annotation and cellular-neighborhood context, not whole-transcriptome pathway projection.

Table S17: **Selected Xenium annotation counts.**

Annotation	Count	Note
Total cells	1,374,915	processed AnnData object loaded in backed mode
Genes	299	targeted panel, not genome-wide
DCT cells	48,774	celltype_plot annotation
Endothelial cells	129,261	celltype_plot annotation
Fibroblasts	160,328	celltype_plot annotation
Injured proximal tubule cells	146,097	celltype_plot annotation
Distal tubule niche cells	77,103	CN annotation
Fibro-inflammatory niche cells	105,536	CN annotation

Interpretation: spatial evidence is a contextual supplement. It supports the wording “spatially contextualized,” not “spatially validated.”

S10 Public perturbation-reference analyses

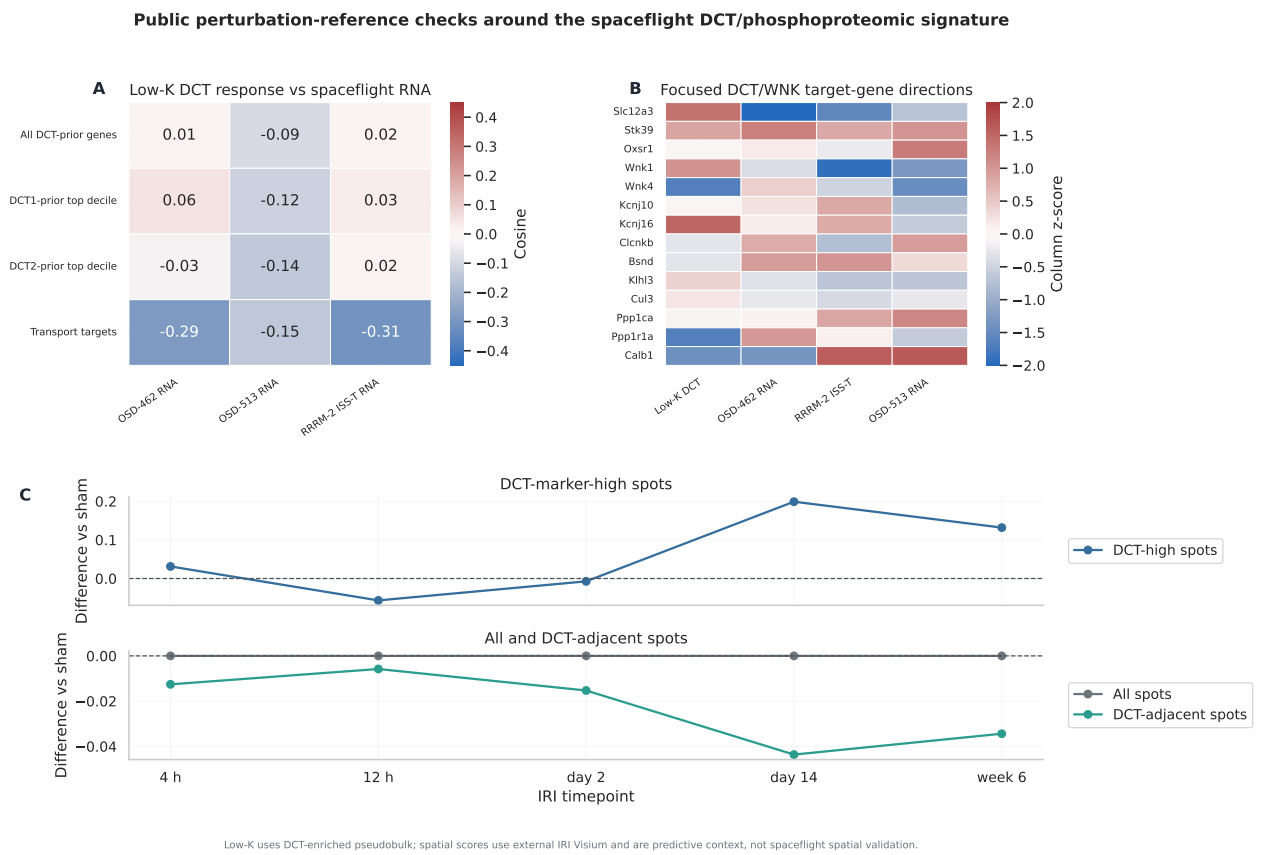


Figure S6: **Public perturbation-reference analyses.** (A) Low-K DCT response cosines with OSD-462, OSD-513, and RRRM-2 ISS-T RNA by gene-prior class. (B) Focused DCT/WNK transport-target gene direction heatmap. (C) Parent-protein-normalized DCT1 top-decile enrichment ORs. (D) IRI Visium DCT-transport score by timepoint and spatial context.

Table S18: **Public perturbation-reference branch verdicts.**

Mechanism branch	Reference and analysis	Headline result	Paper role
Low-K/WNK reference	GSE228367 low-K DCT-enriched pseudobulk anti-alignment	Directional anti-alignment in 14-gene transport-target subset (cosine -0.286 OSD-462, -0.311 RRRM-2 ISS-T, -0.146 OSD-513), wide bootstrap CIs, did not meet promotion rule.	Mechanism triage, not primary upstream mechanism.
Parent-protein normalization	OSD-462 parent-protein-normalized phosphosite re-test	DCT1 top-decile OR 1.52 (CI 1.36–1.70, $q=1.04 \times 10^{-11}$); DCT2-leaning decile also enriched; paired sample-level contrast supportive.	Robustness upgrade to the DCT-subtype-prior result; not direct biochemical occupancy.
Injury/repair spatial context	GSE269622 IRI Visium DCT-transport score by spatial niche	DCT-adjacent spots drop at day 14 (-0.044) and week 6 (-0.034); DCT-marker-high spots rise at the same timepoints.	Spatial prediction/context for future spaceflight kidney sections.
Vasopressin/cAMP	PXD001729 mpkDCT dDAVP	60 shared single sites, 0 shared transport-target sites; key DCT/WNK targets do not map.	Secondary negative; coverage-bounded.
KLHL3/CUL3 turnover	OSD-462 KLHL3/CUL3/WNK coverage	0 <i>Klhl3</i> phospho-sites, 1 <i>Cul3</i> site, flat WNK/SPAK/NCC protein abundance; no public kidney ubiquitinomics.	Future-data rationale.

S10.1 Branch 1: GSE228367 low-K DCT anti-alignment

Table S19: **Low-K DCT response vs spaceflight RNA, 14-gene transport-target subset.**

Spaceflight vector	<i>n</i> genes	Cosine	95% CI	Spearman ρ	<i>p</i>	q
OSD-462 RNA	14	-0.286	-0.752 to 0.458	-0.420	0.135	0.218
RRRM-2 ISS-T RNA	14	-0.311	-0.780 to 0.325	-0.341	0.233	0.306
OSD-513 RNA stat	14	-0.146	-0.768 to 0.506	-0.169	0.563	0.695

Interpretation: the 14-gene transport-target subset moves directionally opposite the canonical low-K DCT activation direction in all three spaceflight cohorts (10 of 14 focused DCT/WNK genes flip in OSD-462, including *Slc12a3*, *Wnk1*, *Klhl3*, *Cul3*), but none of the comparisons clears the pre-specified primary threshold (cosine < -0.3 with CI excluding zero). Promotion rule not met; treated as mechanism triage.

S10.2 Branch 2: parent-protein-normalized DCT-prior enrichment

Table S20: **Parent-protein-normalized phosphosite enrichment among DCT-prior parent genes.** Parent-normalized effect = phosphosite flight effect minus parent-protein flight effect. This is a robustness summary, not direct biochemical occupancy.

Analysis	DCT-prior subset	OR	95% CI	q	Read
parent-normalized $p < 0.05$	DCT1 top decile	1.52	1.36–1.70	1.04×10^{-11}	robustness upgrade holds
parent-normalized $p < 0.05$	DCT2-leaning decile	1.32	1.18–1.48	4.27×10^{-6}	comparator also enriched
parent-normalized strict $q < 0.10$	DCT1 top decile	1.61	1.42–1.82	1.04×10^{-11}	stricter set, OR rises
parent-normalized strict $q < 0.10$	DCT2-leaning decile	1.28	1.13–1.46	3.01×10^{-4}	comparator also enriched
exclude anchor genes	DCT1 top decile	1.51	1.34–1.69	2.73×10^{-11}	survives anchor exclusion
exclude NCC sites	DCT1 top decile	1.51	1.35–1.69	2.73×10^{-11}	survives NCC exclusion
one row per gene	DCT1 top decile	1.50	1.19–1.90	0.00101	survives parent-gene representative-row sensitivity
single-position row per gene	DCT1 top decile	1.48	1.17–1.88	0.00147	stricter representative-row sensitivity passes
paired sample-level contrast	DCT1 top decile	1.56	1.37–1.77	1.90×10^{-10}	matched animal-level contrast supportive
paired sample-level contrast	DCT2-leaning decile	1.54	1.36–1.75	2.22×10^{-10}	matched animal-level comparator supportive

Table S21: **Selected target-site parent-protein-normalized examples.**

Gene	Site	Raw phospho effect	Parent protein effect	Parent-normalized effect
<i>Slc12a3</i> /NCC	p53	−0.851	+0.089	−0.940
<i>Slc12a3</i> /NCC	p65	−0.790	+0.089	−0.879
<i>Slc12a3</i> /NCC	p68	−0.794	+0.089	−0.883
<i>Slc12a3</i> /NCC	p89	−0.930	+0.089	−1.019
<i>Stk39</i> /SPAK	p366	−0.520	+0.081	−0.601
<i>Stk39</i> /SPAK	p383	−0.793	+0.081	−0.874

Interpretation: per-site parent-protein normalization sharpens the suppressed-site effect because parent NCC and SPAK protein effects are slightly positive. The DCT-prior enrichment is not eliminated by parent protein abundance in this sensitivity; the calculation remains a whole-kidney parent-normalized effect rather than direct phosphosite occupancy or biochemical stoichiometry.

S10.3 Branch 3: IRI Visium DCT-transport spatial context

Table S22: **Visium DCT-transport score versus sham, by spatial group and timepoint.** Spot-level descriptive two-sample t ; no animal-level replication.

Spatial group	Timepoint	Mean diff vs sham	t	p	n spots vs sham
DCT-adjacent spots	day 14	−0.044	−2.81	0.0050	932 vs 1,134
DCT-adjacent spots	week 6	−0.034	−2.36	0.0182	1,574 vs 1,134
DCT-marker-high spots (top quartile)	day 14	+0.199	+8.02	2.25×10^{-15}	693 vs 715
DCT-marker-high spots (top quartile)	week 6	+0.132	+5.98	2.70×10^{-9}	979 vs 715

Interpretation: DCT-adjacent neighborhoods carry a DCT-transport drop in late IRI repair while DCT-marker-high spots retain or even increase the DCT-transport program. This is the spatial prediction that a future spaceflight kidney spatial experiment would test directly.

S10.4 Branch 4: PXD001729 dDAVP/mpkDCT

Table S23: PXD001729 mpkDCT dDAVP reference vs OSD-462.

Quantity	Value	Note
Shared single phosphosites	60	cross-platform/cross-species mapping
Shared changed transport-target sites	0	<i>Slc12a3</i> , <i>Stk39</i> , <i>Oxsr1</i> , <i>Wnk1</i> , <i>Wnk4</i> , <i>Klhl3</i> , <i>Cul3</i> not shared
All-shared-site cosine	−0.026	95% bootstrap CI −0.310 to 0.287

Interpretation: the cultured DCT-lineage dDAVP reference is coverage-bounded for the OSD-462 transport-target lesion and cannot test a vasopressin/cAMP anti-alignment hypothesis. Used as plausibility/coverage boundary only.

S10.5 Branch 5: KLHL3/CUL3/WNK turnover

Table S24: KLHL3/CUL3/WNK coverage in OSD-462 and PXD001729.

Gene	OSD-462 phosphosites	PXD phosphosites	OSD-462 protein effect	Read
<i>Klhl3</i>	0	0	+0.013	no phosphosite evidence
<i>Cul3</i>	1	0	−0.001	no turnover signal
<i>Wnk1</i>	12	12	−0.024	phospho observed, protein flat
<i>Wnk4</i>	8	1	+0.087	phospho observed, protein not depleted
<i>Stk39</i> /SPAK	5	3	+0.081	phospho suppressed despite flat protein
<i>Slc12a3</i> /NCC	16	0	+0.089	phospho suppressed with protein flat/up

Interpretation: public data cannot distinguish KLHL3/CUL3-mediated WNK turnover from ionic/osmotic suppression. The next experiment requires DCT-enriched or targeted ubiquitinomics and direct WNK protein turnover assays.

S11 Human, aldosterone, and CMap appendix context

These analyses are not used as validation of the mouse kidney phosphoproteomic result. They document human-facing fluid-axis context, a directional mineralocorticoid hypothesis, and an approximate perturbagen-class screen.

Table S25: Human urine/fluid-axis context and OSD-656 recovery-panel inventory.

Analysis	Headline value	Interpretation
Twins urine/fluid-axis concordance	Three physiological axes concordant; exact two-sided sign test $p = 0.25$	Directional single-flight-subject context only; not human kidney-omics validation.
Figure-level <i>AQP2</i> direction	Up in Supplementary Fig. S4A	Encoded as direction only; not machine-readable kidney tissue evidence.
OSD-656 Inspiration4 urine panel	203 summarized markers; 92 inflammation/matrix-context markers; 0 direct distal-nephron/RAAS markers in predefined category	Post-flight recovery context; not independent in-flight distal-nephron validation.

Table S26: **Appendix mechanism-context screens.**

Analysis	Headline value	Interpretation
Mineralocorticoid/aldosterone axis	Meta-effect -0.840 , $p = 0.060$	Directional hypothesis-generating context; below the declared significance threshold.
LINCS/CMap local screen	50 up and 50 down BING-compatible query genes; 473,647 Level-5 signatures scored after <code>sig\info</code> merge	Approximate cell-line perturbagen-class screen; not a treatment or countermeasure analysis.
Ortholog/symbol mapping	Uppercase mouse-to-human matching unless a dedicated ortholog table is supplied	Mapping limitation; candidate lists should not be over-interpreted.

Source artifacts:

- `human_concordance/human_concordance_summary.tsv`
- `human_concordance/twins_axis_level_concordance.tsv`
- `human_concordance/twins_aqp2_figure_evidence.tsv`
- `human_concordance/osd656_prepost_summary.tsv`
- `regulator_activity/aldosterone_axis_summary.tsv`
- `cmap_screen/cmap_summary.tsv`

S12 Complete artifact index

Table S27: **V11 generated result artifacts indexed by source path.**

Category	Source artifact(s)
Run report	<code>docs/v11_execution_research_plan.md</code> ; <code>docs/v11_execution_results.md</code>
Headline-number index	<code>latex_paper/v11_headline_number_index.tsv</code> ; maps reported manuscript numbers to source artifacts, filters, and columns.
V11 source modules	<code>src/v11/build_gse228367_dct_prior.R</code> ; <code>src/v11/core_analysis.py</code> ; <code>src/v11/h2_composition_aware_phospho.py</code> ; <code>src/v11/channel_centering_sensitivity.py</code> ; <code>src/v11/kinome_atlas_ksea.py</code> ; <code>src/v11/recurrence_meta.py</code> ; <code>src/v11/rna_protein_propagation.py</code> ; <code>src/v11/observability_audit.py</code> ; <code>src/v11/spatial_reference_projection.py</code> ; <code>src/v11/publication_figures.py</code> ; <code>src/v11/perturbation_gse228367_lowk.py</code> ; <code>src/v11/h2_occupancy_normalized_phospho.py</code> ; <code>src/v11/spatial_dct_transport_check.py</code> ; <code>src/v11/perturbation_triage_summary.py</code>
Runner integration	<code>src/run_all_phases.py</code> <code>phase 11</code> ; <code>--v11</code> ; <code>--v11-only</code> ; <code>--v11-site-scope</code> ; <code>--v11-skip-spatial</code> ; <code>--v11-skip-visium</code> ; <code>--v11-skip-xenium</code> ; <code>--v11-skip-figures</code>
Baseline lock	<code>data/results/run_20260526_v11_dct1_phospho_mediation/baseline/v11_baseline_lock_summary.tsv</code> ; <code>baseline/v11_baseline_input_manifest.json</code> ; <code>baseline/osd462_targeted_ksea_substrates.tsv</code> ; <code>baseline/rna_recurrence_gene_set_members.tsv</code> ; <code>baseline/osd462_rna_recurrence_leave_one_family.tsv</code> ; <code>baseline/osd462_rna_recurrence_paired_pathway_bootstrap.tsv</code> ; <code>baseline/osd462_tmt_channel_qc.tsv</code> ; <code>baseline/osd462_tmt_missingness_by_condition.tsv</code>
Cross-layer pathway matrix	<code>cross_layer/osd462_cross_layer_pathway_matrix.tsv</code> ; <code>cross_layer/osd462_cross_layer_pattern_summary.tsv</code> ; descriptive RNA/protein/phosphosite pathway-layer summary generated by <code>src/v11/core_analysis.py</code> .

Category	Source artifact(s)
Calibrated RNA→protein propagation	run_20260606_v11_layer_specificity/propagation/rna_protein_propagation_summary.tsv; propagation/rna_protein_propagation_gene_classes.tsv; propagation/rna_protein_propagation_coverage.tsv; figures/v11_rna_protein_propagation.pdf; matched-null per-pathway layer assignments from src/v11/rna_protein_propagation.py.
Proteome observability audit	run_20260606_v11_layer_specificity/observability/v11_observability_per_gene.tsv; observability/v11_observability_detectability_gradient.tsv; observability/v11_observability_matched_propagation.tsv; observability/v11_observability_q_delta.tsv; observability/v11_observability_high_coverage_propagation.tsv; observability/v11_observability_ncc_site_audit.tsv; detectability-bias control from src/v11/observability_audit.py.
DCT reference prior	dct_prior/gse228367_gene_stats_overall.tsv; dct_prior/gse228367_gene_stats_by_rep.tsv; dct_prior/gse228367_dct1_vs_dct2_de.tsv; dct_prior/gse228367_dct_prior_summary.json; dct_prior/gse228367_dct_prior_score_sensitivity.tsv; dct_prior/gse228367_anchor_gene_prior_positions.tsv; dct_prior/gse228367_prior_bin_stability.tsv
External QC	external_qc/gse228367_object_inventory.tsv; external_qc/gse228367_marker_qc.tsv; external_qc/pxd001729_table_inventory.tsv; external_qc/pxd001729_phosphosite_effects.tsv
DCT-subtype-prior enrichment	h2_enrichment/h2_dct1_sensitivity_summary.tsv; h2_enrichment/h2_dct1_parent_gene_level_summary.tsv; h2_enrichment/h2_dct1_site_count_stratified_permutation.tsv; h2_enrichment/h2_dct_extreme_bin_cluster_bootstrap.tsv; h2_enrichment/h2_dct_matched_parent_gene_permutation.tsv; h2_enrichment/h2_dct_threshold_free_summary.tsv; h2_enrichment/h2_dct_percentile_threshold_sensitivity.tsv; h2_enrichment/h2_dct_channel_centering_sensitivity.tsv; h2_enrichment/h2_dct_channel_centering_effect_comparison.tsv; h2_enrichment/h2_dct_channel_centering_verdict.json; h2_enrichment/h2_dct1_enrichment_verdict.json
Kinome and recurrence extensions	regulator_activity/osd462_kinome_atlas_ksea.tsv; regulator_activity/osd462_kinome_atlas_net_summary.tsv; cross_osdr_recurrence/recurrence_meta_summary.tsv; cross_osdr_recurrence/recurrence_meta_gene_set_scores.tsv; cross_osdr_recurrence/recurrence_meta_verdict.json
PXD001729 checks	h2_pxd/h2_pxd001729_ddavp_antialignment_summary.tsv; h2_pxd/h2_pxd001729_shared_sites.tsv; h2_pxd/h2_pxd001729_verdict.json
KLHL3/CUL3 checks	h2_klh13/h2_klh13_cul3_effects.tsv; h2_klh13/h2_klh13_cul3_verdict.json
Composition-aware models	h2_composition_adjusted/h2_composition_adjusted_suppression_enrichment_single.tsv; h2_composition_adjusted/h2_composition_effect_level_dct1_ladder_single.tsv; h2_composition_adjusted/h2_composition_site_fixed_interaction_ladder_single.tsv; h2_composition_adjusted/h2_composition_fixed_set_ladder_single.tsv; h2_composition_adjusted/h2_composition_covariate_diagnostics_single.tsv; h2_composition_adjusted/h2_composition_aware_verdict_single.json; h2_composition_adjusted/h2_composition_manifest_single.json

Category	Source artifact(s)
Exploratory covariance decomposition	h3_mediation/h3_mediation_model_summary.tsv; h3_mediation/h3_mediation_posterior_draws.tsv.gz; h3_mediation/h3_mediation_power_simulation.tsv; h3_mediation/h3_mediation_verdict.json
Spatial reference	spatial_reference/spatial_reference_projection_verdict.json; spatial_reference/visium_timepoint_gene_cosines.tsv; spatial_reference/visium_timepoint_pathway_cosines.tsv; spatial_reference/visium_niche_cosines.tsv; spatial_reference/visium_spot_niche_counts.tsv; spatial_reference/xenium_annotation_inventory.tsv; spatial_reference/visium_dct_transport_by_niche.tsv; spatial_reference/visium_dct_transport_timepoint_summary.tsv; spatial_reference/visium_dct_transport_spot_scores.tsv.gz; spatial_reference/visium_dct_transport_verdict.json
Perturbation triangulation	perturbation/lowk_dct_pseudobulk_effects.tsv; perturbation/lowk_dct_alignment_summary.tsv; perturbation/lowk_target_gene_table.tsv; perturbation/lowk_dct1_dct2_specificity.tsv; perturbation/lowk_alignment_verdict.json; perturbation/perturbation_triage_summary.tsv; perturbation/perturbation_triage_verdict.json
Parent-protein-normalized re-test	h2_occupancy/h2_occupancy_site_effects.tsv; h2_occupancy/h2_occupancy_dct1_enrichment.tsv; h2_occupancy/h2_occupancy_target_sites.tsv; h2_occupancy/h2_parent_normalized_paired_site_effects.tsv.gz; h2_occupancy/h2_parent_normalized_paired_enrichment.tsv; h2_occupancy/h2_occupancy_verdict.json
Human context	human_concordance/human_concordance_summary.tsv; human_concordance/human_concordance_verdict.json; human_concordance/twins_axis_concordance.tsv; human_concordance/twins_axis_level_concordance.tsv; human_concordance/twins_aqp2_figure_evidence.tsv; human_concordance/osd656_prepost_summary.tsv
Aldosterone and CMap appendix	regulator_activity/aldosterone_axis_summary.tsv; regulator_activity/aldosterone_axis_verdict.json; cmap_screen/cmap_summary.tsv; cmap_screen/query_genes.tsv; cmap_screen/signature_connectivity.tsv; cmap_screen/perturbagen_summary.tsv; cmap_screen/moa_summary.tsv
V11 figures	figures/v11/v11_dct1_vs_dct2_evidence_ladder.pdf; figures/v11/v11_tmt_channel_centering_qc.pdf; figures/v11/v11_extension_kinome_recurrence.pdf; figures/v11/v11_dct1_parent_gene_enrichment.pdf; figures/v11/v11_distal_nephron_prior_enrichment.pdf; figures/v11/v11_parent_protein_composition_sensitivity.pdf; figures/v11/v11_exploratory_mediation_forest.pdf; figures/v11/v11_exploratory_covariance_decomposition.pdf; figures/v11/v11_spatial_reference_projection.pdf; figures/v11/v11_xenium_annotation_inventory.pdf; figures/v11/v11_perturbation_triangulation.pdf
Downloaded spatial data	data/external/spatial_reference/GSE269622_Visium/*.tar.gz; data/external/spatial_reference/GSE269719_Xenium/Xenium.h5ad

S13 Proteome observability-bias audit

Detectability-bias control for the cross-layer propagation pattern. Per-gene observability features (`n_peptides`, `abundance_log2`, and a peptide-weighted `missing_fraction`) were collapsed from `protein_effects_by_row.tsv`, and the per-pathway matched-null tests behind Table S6 were re-run with the `abundance×peptide`

null extended to a $5 \times 4 \times 3$ abundance $\times$ peptide $\times$ missing-fraction stratum.

Detectability gradient. Fraction of RNA-detected genes also protein-quantified, by |RNA effect| decile: 0.468, 0.489, 0.478, 0.445, 0.387 (top decile). The relationship is roughly flat with a ~ 10 p.p. dip in the top decile—not steep enough to explain the 7-of-11-pathway discordance.

Observability-matched re-test. Per-pathway $|\Delta \text{BH } q|$ between the standard and observability-extended stratifications is at most 0.041 (full table: `observability/v11_observability_q_delta.tsv`). Adding miss- ingness to the null does not shift any layer assignment.

High-coverage subset. Restriction to $n_{\text{peptides}} \geq 3$ and $\text{missing_fraction} \leq 0.20$ retains 6,282/6,549 = 95.9% of the pool, and the propagation pattern reproduces on the subset (`observability/v11_observability_ high_coverage_propagation.tsv`).

NCC/SPAK phosphosite observability. The five pre-specified NCC/SPAK regulatory phosphosites all sit in the top 0.9% of phosphoproteome effect magnitude, while the four non-regulatory NCC control sites sit at the 15–28th percentile—the suppression is concentrated in the most-confidently-measured sites, not in a low-coverage tail.

Gene	Site	Role	Coverage pct	effect pct
Slc12a3	S53	regulatory	66.7	99.37
Slc12a3	S65	regulatory	66.7	99.15
Slc12a3	S68	regulatory	66.7	99.18
Stk39	S382	regulatory	16.7	99.29
Stk39	S383	regulatory	66.7	99.18
Slc12a3	S96	non-regulatory control	66.7	26.07
Slc12a3	S120	non-regulatory control	66.7	14.71
Slc12a3	S122	non-regulatory control	66.7	24.23
Slc12a3	S124	non-regulatory control	66.7	27.62

S14 Verification notes

- The v11 figure module generated PNG/PDF figure sets under `figures/v11/`, including the DCT1-vs-DCT2 evidence ladder, TMT channel-centering QC, kinome/recurrence extension panel, distal-nephron prior enrichment alias, parent-protein composition sensitivity, exploratory covariance-decomposition alias, spatial-reference projection, Xenium annotation inventory, and perturbation triangulation.
- The direct v11 figure generator and integrated `src.visualization.publication_figures` path both detect and regenerate v11 figures.
- Python syntax checks passed for the v11 core, composition-aware, spatial-reference, figure, perturbation low-K, parent-protein-normalized phosphosite, spatial DCT-transport, and triage modules, plus the runner and publication-figure integration.
- The runner dry-run passed for `python -m src.run_all_phases --v11-only --dry-run --run-id run\_20260526\_v11\_dryrun --v11-skip-spatial` and for the perturbation-triangulation modules.
- The GSE269719 Xenium MD5 matched the figshare record: `e6f5728ab7e47b499149f63b54e892c7`.
- The spatial scripts completed after loading six GSE269622 Visium archives and the processed GSE269719 Xenium AnnData object; the DCT-transport spatial check used the same Visium archives.
- The perturbation low-K analysis loaded six GSE228367 filtered 10x matrices (NK1–3 and KD1–3) and emitted alignment and target-gene tables under `perturbation/`.

Curated results compendium for Manuscript v11. This document should travel with `manuscript_v11.tex` and `manuscript_v11.pdf`. The manuscript result is a matched RNA/protein/phosphoprotein constraint map around the established spaceflight kidney NCC/SPAK/WNK phospho-phenotype, with distal-nephron subtype-prior enrichment as a bounded secondary prioritization layer.